

# Iris Yang

<https://irisxyang.github.io>

iris.xy.yang@gmail.com • 100 Memorial Dr Apt 203A, Cambridge, MA 02142 • 425-922-7730

## EDUCATION

**Massachusetts Institute of Technology** | Cambridge, MA

*Exp. May 2025*

GPA 4.5/5.0 | *Candidate for B.S. in Computer Science and Business Analytics*

## SKILLS & PROJECTS

**Skills:** Python, C, Java, Typescript, JS/HTML/CSS, React, Node, R, Sawtooth, PyTorch, Data Analysis, Data Structures

**Relevant Coursework:** Software Construction/Design, Computational Thinking/Data Science, Algorithm Design/Analysis, Intro to C, Statistics, Linear Algebra, Math for CS, Machine Learning, Natural Language Processing

**Paper: Enhancements on Beam Search in LLMs**

*December 2024*

- Sought to reduce hallucinations in model output via leveraging attention scores as a metric for uncertainty for dynamic beam sizing
- Implemented and tested modifications to beam search using GPT-2 as a base model in PyTorch

- Link: <https://github.com/irisxyang/beam-search-paper>

**Multimodal Handwriting Text Recognition/Transcription Pipeline**

*April 2024*

- Implemented a full handwriting text recognition pipeline using Google Cloud Vision API in Python
- Improved text detection performance by implementing various image processing optimization methods
- Created a usable web interface for the pipeline using Python Flask

- Github link: <https://github.com/irisxyang/htr-proj>

**PixelThis Web App** | Cambridge, MA

*January 2023*

- Designed and built a full-stack web app from scratch using JavaScript, Node.js, React.js, and Express
- Implemented interactive elements on the site such as a useable canvas, user sign-in, and posting capabilities

- Link: <https://pixel-this.herokuapp.com/>

**Morgan Stanley 2022 Code To Give Hackathon – 2nd Place Winner**

*March 2022*

- Developed a front-end website interface in a team of 4 to help aid those facing homelessness

## EXPERIENCE

**Data Monetization Analysis – MIT CISR** | Cambridge, MA

*May 2024–Jan 2025*

*Data Science Researcher*

- Created Python data processing pipeline to collect, parse, and clean input data to be used for further analysis
- Performed statistical analyses on input data to adjust for outliers and generalize results
- Helped formulate processes to determine industry benchmark statistics based on financial data from LSEG

**ML Data Extraction Pipeline Implementation – MIT CSAIL** | Seattle, WA

*May 2023–July 2023*

*Machine Learning Researcher*

- Created frameworks in Python to parse product data from electronic product specification documents
- Implemented ML models to help build a semi-automated extraction pipeline for data extraction

**LLM Model Performance on College Course Material** | Virtual

*April 2023–July 2023*

*Artificial Intelligence Researcher*

- Researched ability of fine-tuned LLMs such as LLaMA in zero-shot learning of MIT Courses
- Verified model output using GPT-4 to compare and analyze model performance
- Link to research: <https://bit.ly/3qyxYIC>

**Store Closure Optimization Model – MIT Media Lab** | Cambridge, MA

*May 2022–Sept 2022*

*Economics and Data Science Researcher*

- Filtered mobile tracking data to calibrate input for a modified Huff Model using Python
- Created and conducted tests to determine the optimal store closure scenarios to maximize profit

## INVOLVEMENT, SKILLS, AND INTERESTS

**Languages:** English and Chinese (Fluent), French (Limited Proficiencies)

**Activities:** MIT Varsity Swimming, Society of Women Engineers at MIT, MIT 2025 Ring Committee

**Interests:** Visual Arts (Portfolio: <http://irisxyang.github.io/portfolio>), Lifting, Coffee